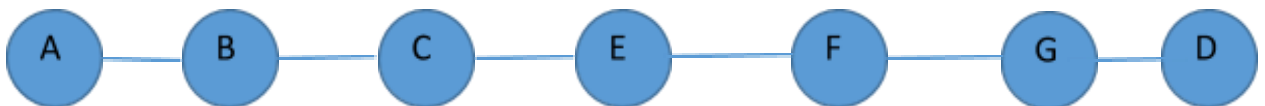


Wireless Networks (CSE538/ECE538)

Assignment 2

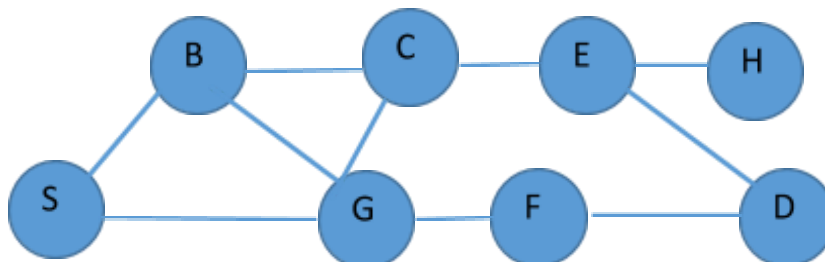
- Consider this multihop network of 6 nodes. Each node is within radio range of its adjacent node. Every node has a half-duplex radio, that can only send or receive (but not both) at a time.



A has an interference range of 2
 B has an interference range of 3
 C has an interference range of 1
 E has an interference range of 4
 F has an interference range of 1
 G has an interference range of 1
 D has an interference range of 1

- Now, calculate the maximum transport datarate (/throughput) of all the nodes A-G when the maximum possible transport datarate is R (your answer should be in terms of R) [4]
- Now, calculate the maximum transport datarate (/throughput) of node A-G according to PIP protocol. Assume there are 3 non overlapping channels (i.e., transmissions over these 3 channels won't interfere with each other). Show the slot and channel assignment of all the nodes. Compute the max possible transport datarate (or throughput) of the network (your answer should be in terms of R) [4]

- Consider the following multihop topology. A link between node A — B means, they are in radio range of each other.



The cost of each link is 1. Assume that D is the only possible destination in the routing and forwarding tables. The nodes run DSDV, DSR and AODV routing protocol one at a time. Answer the following questions.

For DSDV and AODV: For this destination D, each node maintains a set of routes, which are tuples of the form (Destination, next-hop, metric, sequence number). The node then picks one of these routes as its best route, and periodically advertises it to neighbors using DSDV-like routing messages. List all known routes (next-hop, metric) to the destination D at nodes B, E, G and F. Also indicate which of these is the best route used by the node for forwarding. You may provide the answers assuming the routing and forwarding information has converged to a steady state after any changes.

For DSR and AODV: show the path followed by RREQ and RREP packets. Show the best path chosen by these two routing protocols.

[3+3]+[3+3]

3. In the PIP paper the EOF is sent after the data transmission. Explain if there are (any) issues if EOF is not sent. Therefore, SNACK is sent after receiving the data. [2]

3. Consider a multihop network of 6 nodes

N1-N2-N3-N4-N5-N6.

Node N1 is the source, and N6 is the destination. The source, N1, broadcasts the first batch of 10 packets with the forwarder list set to N6, N5, N4, N3, N2 (ordered from highest to lowest priority). Assume that the packets contain only 1 byte and they are A,B,C,D,E,F,G,H,I,J. The reception ratio for the packets sent from N1 are 0%, 10%, 20%, 40%, 60% for the nodes N6, N5, N4, N3, N2 respectively. So, out of these 10 packets, N6 did not receive any packet, N5 1, N4 2, N3 4, N2 6 packets. Assume that the nodes have received the following set of packets N5 => G, N4 => G, H, N3=> E, F, G, H, N2=> A,B,E,F,G,H. Also, assume that N6, N5, N4, N3 can hear themselves well (100% success rate). N6, N5, N4, can't hear N2. N3 and N2 can hear themselves well (100% success rate). Use ExOR opportunistic routing protocol to route the packets till the destination receives atleast 90% of the packets. How many rounds from the source would be needed? Show the timing diagram of the ExOR transfers. [6]